

Leonardo Santos

Electrical Engineer | Data Scientist | Software Developer

Electrical engineer with experience in electrical projects, integrated circuits, firmware, and software. Specializing in digital systems for signal processing and data science, with a focus on innovation and efficiency.

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🌐 je-leo-as.github.io

ACADEMIC BACKGROUND

Bachelor's Degree in Electrical Engineering

Federal University of Paraná

03/2019 - 12/2024

Courses

- Monitor of Digital Electronics and Microelectronics, assisting with exercises, questions, and lab practices.
- Taught Python course for Chemical Engineering.
- Administrative manager of Seatel, organizing courses, lectures, and technical visits.
- Participant in PET: IoPET (IoT and firmware), BigPET (machine learning, leader), PET3D (design and 3D printing, leader).
- Developed electrical subsystem for off-road vehicle (wiring harness, PCB, firmware, 3D modeling in Catia). Achievements: 1st place in Regional South electrical project 2019, 11th place nationally Baja SAE 2020, 2nd place in Regional South electrical project 2020.

Master's Degree in Electrical Engineering

Federal University of Paraná

03/2025 - Present

Courses

- Research in digital circuits for telecommunications, focusing on signal processing and hardware optimization.
- Study of FPGA and ASIC architectures for energy efficiency in embedded wireless communication systems.

PROFESSIONAL EXPERIENCE

Software Developer / Data Scientist

IoTag Tecnologia

10/2022 - 02/2026

Company focused on precision agriculture

Achievements/Tasks

- Developed firmware in C/C++ for virtual terminal interfaces for agricultural machinery.
- Created data science algorithms for spatial data and equipment collection.
- Implemented scalable cloud services with AWS and databases.

Engineering Intern

Lactec

06/2022 - 10/2022

Technology and research institute

Achievements/Tasks

- Assisted in the development of capacitive water level and flow sensors for tidal energy generators.
- Enhanced skills in electronics, microcontroller programming, and system integration.

COMPLEMENTARY ACTIVITIES

Member of Electrical Subsystem and Management

UFPR Baja SAE

01/2019 - 12/2021

Extension project for off-road vehicle

Achievements/Tasks

- Developed wiring harness, PCBs, and firmware.
- 3D modeling in Catia.
- Achievements: 1st place in Regional South electrical project 2019, 11th place nationally 2020, 2nd place in Regional South 2020.

Scientific Researcher

GICS

08/2021 - 12/2024

Research group in circuits and systems

Achievements/Tasks

- Developed TRNG in VHDL, implemented in FPGA and designed with Cadence.
- Modeled DPD based on Memory Polynomial in Python, implemented in VHDL and validated with Xilinx.

Project Manager and Member

Tutorial Education Program (PET)

08/2021 - 12/2024

Undergraduate enhancement program

Achievements/Tasks

- IoPET: Developed electronic lock with ESP01 and MQTT.
- BigPET: Led machine learning projects, taught course and developed fake news detector.
- PET3D: Led design and 3D printing, including supports for face shield masks during the pandemic.
- Monitor of Digital Electronics and Microelectronics.

PROJECTS

Virtual Assistant

- Multimodal assistant with LangChain for NLP, FastAPI for backend, and speech recognition. Frontend in JavaScript.

Electronic Lock

- Smart lock with ESP-01 (ESP8266) and MQTT protocol for remote control.

TRNG Integrated Circuit

- TRNG in 130 nm with VHDL, optimized in FPGA, synthesized with Genus and routed with Innovus (43 cells, 41 μW).

Fake News Detector

- System for verifying the truthfulness of news using text processing and machine learning (Logistic Regression, KNN, Random Forest).

Automatic Resume Translator

- Automated resume translation system with CrewAI and Typst, hosted on GitHub (Je-Leo-AS/Curriculo), with versions in Portuguese (PT-BR) and English (EN) in separate branches. Implemented GitHub Actions to trigger automatic translations via CrewAI, updating the EN branch.

n8n Automation — AI agents for maintenance and communication management

- Automation platform with n8n to orchestrate workflows and internal AI agents.
- Agent “Proxmox Care”: monitors the Proxmox server on the private network and performs maintenance tasks (resource checks, anomaly notifications, scheduling updates, backup, and cleanup routines). Operates in an isolated environment with key authentication and restricted permissions to preserve network security.
- Agent “Comms Manager”: manages messages and emails from the professional number and email — ingests, classifies by priority, automatic summaries, standard responses, and routing to channels (tickets, Slack, CRM). Uses NLP to detect urgency and extract relevant entities.
- Integration via webhooks, queues, and audit logs; automations triggered by events, with logs and notifications to ensure traceability and human review when necessary.

SKILLS

Python

VHDL

C/C++

Java

JavaScript

PHP

HTML/CSS

3D Modeling (SolidWorks, Fusion 360, Catia)

Circuit Design (Cadence, ADS)

Circuit Simulation (Proteus)

PCB (Altium Designer)

Python Frameworks (OpenCV, FastAPI, LangChain)

Databases and SQL

Web Development

Interpersonal Communication

Planning

Time Management

N8N

CERTIFICATIONS

Android Development

60 hours

Machine Learning

60 hours

Databases

60 hours

LANGUAGES

Portuguese (native)

English (C1)